

Balancing the Scale: Solving Equations Visually

PhET Equality Explorer · PhET

Years 7-10

~30 minutes

Mathematics · Algebra

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/equality-explorer>

Curriculum links: AC9M7A01 · AC9M7A02 · AC9M10A01 · AC9M10A02

Learning intentions

- I can represent an unknown quantity with a variable and set up an equation as items on the balance scale.
- I can use inverse operations to keep the scale balanced while solving for the unknown.
- I can solve a linear equation algebraically and check my answer using the scale.

Getting started

Open Equality Explorer and start on the Basics or Numbers screen. Build a simple equation using the items provided, keeping both sides of the scale balanced as you add or remove pieces. Once you're comfortable, move to a screen with variables (x) for the later tasks.

Tasks

1. Build an equation using numbers only, so that both sides of the scale balance. Write down the equation you built.

2. On a variables screen, set up an equation such as $x + 3 = 7$ as items on the scale. Describe the single action you can take to isolate x while keeping the scale balanced.

3. Solve the equation from the previous task algebraically, writing each step. Then verify your answer by removing or adding matching items on both sides of the scale.

Working:

4. Explain why you must always do the same thing to both sides of the scale, and connect this to why we do the same operation to both sides of an equation algebraically.

5. Set up an equation with x on both sides, such as $2x + 1 = x + 5$. Solve it using the scale, showing your working.

Working:

6. Try three different equations with larger or negative coefficients. Record each starting equation and its solution.

Equation	Solution ($x = ?$)

7. Describe one step in solving an equation that was harder to represent with physical items on the scale (for example, negative numbers or fractions), and explain how you worked around it.

Extension (optional)

Use Equality Explorer to model a real situation as an equation — for example, the cost of a taxi ride with a flat fee plus a rate per kilometre. Solve it and explain what your solution means in the real-world context.